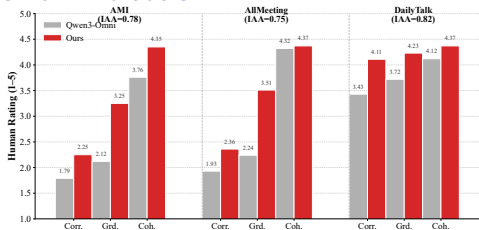


Analysis: Human Evaluation & Ablation Study

Human Evaluation



Blind evaluation on 150 queries (3 annotators, IAA= 0.78).

+0.97 Groundedness confirms fewer hallucinations.

Ablation Study (AMI)

Setting	Avg. %	Δ
Full GRGA	49.49	—
w/o Query Planner	44.68	−4.80
w/o Reflection	39.64	−9.84
w/o Graph Traversal	38.21	−11.27
w/o Audio Access	34.71	−14.78
w/o Semantic Search	16.28	−33.21

Evidence Precision/Recall (AMI, $\pm 2s$)

Method	P	R
Qwen3-Omni	25.7	27.3
+ Text RAG	56.9	65.2
+ GRGA	76.2	69.7